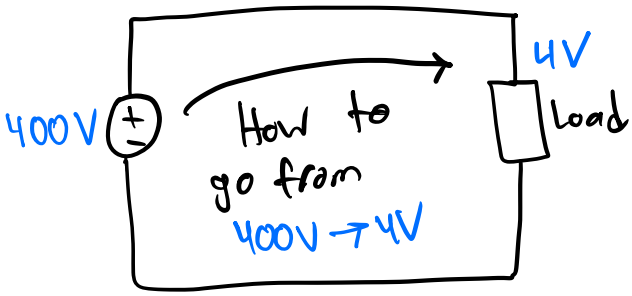
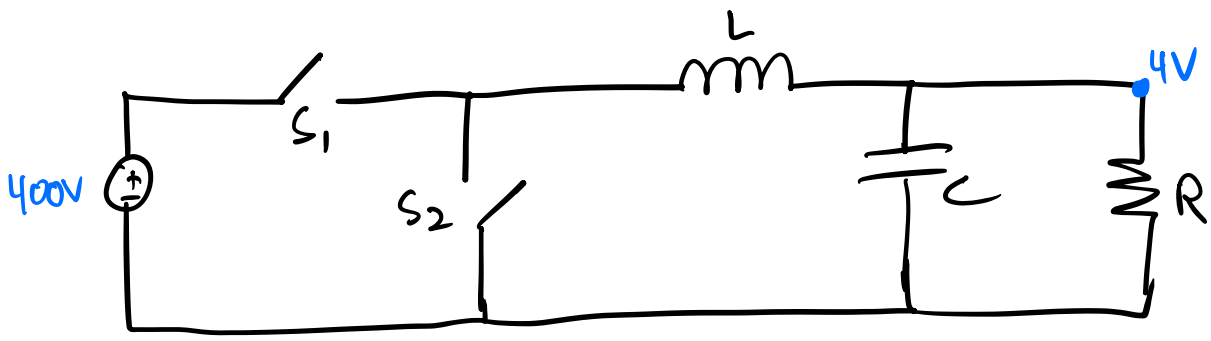


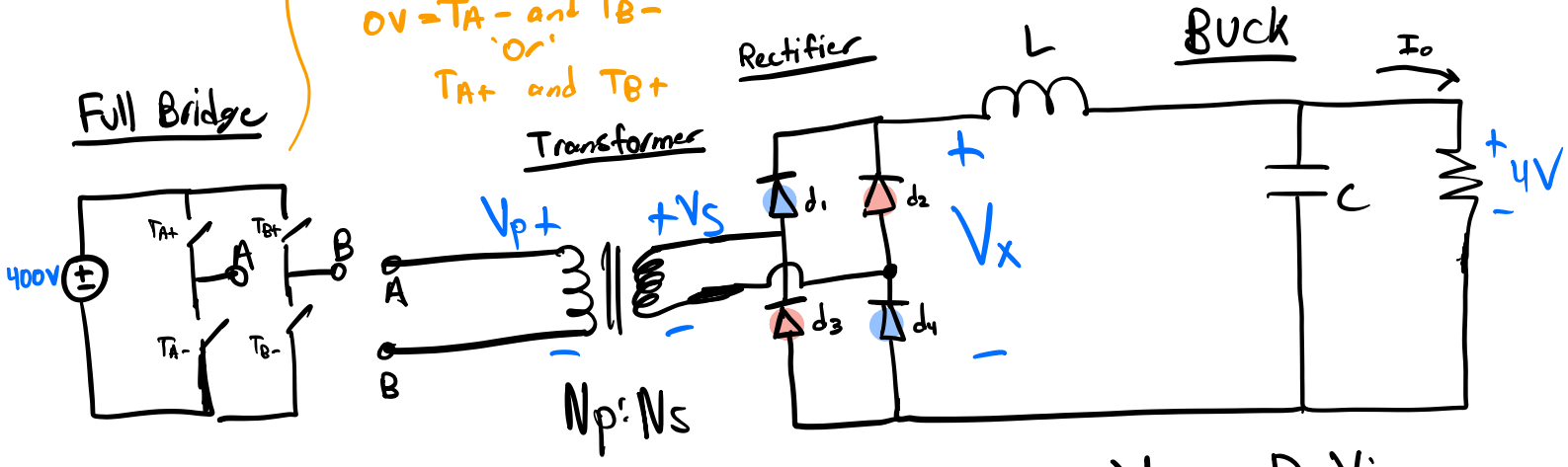
↓

<u>Use a Buck</u> with $D = 0.01$



400V = TA+ and TB−
−400 V = TB+ and TA−
0V = TA− and TB−
'or'
TA+ and TB+

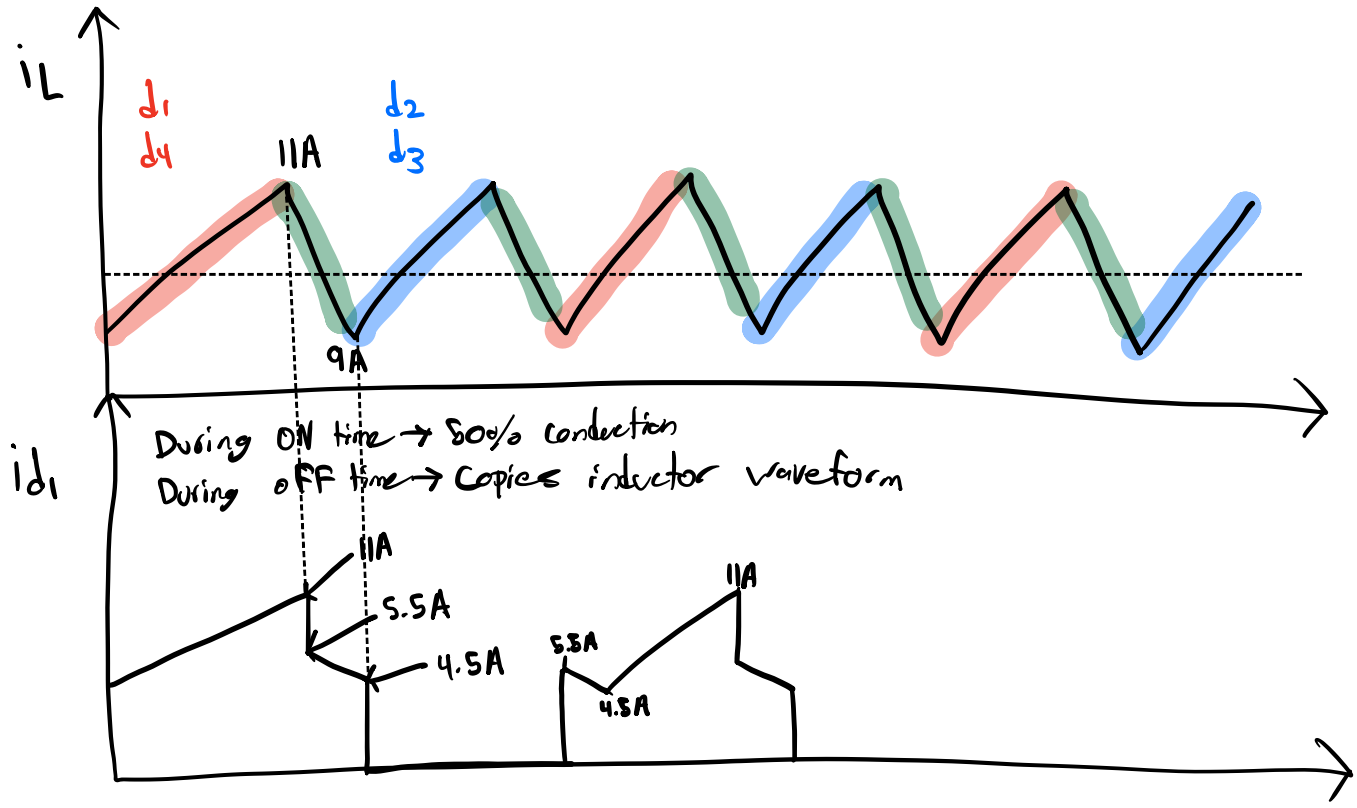
$D = 0.7$



$V_o = D \cdot V_{in}$

$\therefore V_{in} = \frac{4}{0.7} = 5.71V$

$\frac{V_p}{V_s} = \frac{N_p}{N_s} = T \rightarrow \frac{400V}{5.71V} = 70.05$ Turns


i_L
d_1
d_4
d_2
d_3
11A
9A
i_{d_1}
During ON time → 50% conduction
During OFF time → Copies inductor waveform
11A
5.5A
4.5A
5.5A
4.5A
11A